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Probabilistic net load forecasting based on transformer network and Gaussian process-enabled residual modeling learning method

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ABSTRACT

Accurate net load forecasting plays an increasingly pivotal role in ensuring the reliable operation and scheduling of power systems. This paper introduces a novel probabilistic net load forecasting approach that combines the strengths of a Transformer network with Gaussian process regression. The state-of-the-art Transformer network is first employed to capture the net load pattern utilizing relatively abundant historical training samples. The remarkable temporal feature extraction ability allows it to discover the complex structure in net loads. Subsequently, the Gaussian Process is applied to capture the behavior of the Transformer network by modeling its forecasting residual utilizing a specific composite kernel. The modeling of the forecasting residual not only provides valuable uncertainty quantification of net load but also improves the forecasting performance based on the Transformer network. Comparative tests utilizing real-world data verify the superiority of the proposed method over other state-of-the-art net load forecasting algorithms.

1. Introduction

The increasing energy demand, energy security, and carbon emissions from fossil fuel generation are pushing the power system to improve the proportion of renewable energy integration. Distributed generation, such as photovoltaics (PVs), is being installed across all voltage levels, especially in low-voltage (LV) networks [1]. Also, owing to the clean, and economical characteristics of electric vehicles (EVs), they are becoming more and more popular around the world [2]. However, the high penetration rate of variable renewable energy sources and EVs inevitably brings more complexity for system operation planning due to multiple uncertain variables such as load, wind, and solar power generation. The additional intermittent and variability from these renewable energy generations also bring bigger challenges in load forecasting, balancing the power grid, and managing ancillary systems [3].

Accurate load forecasting can contribute to the reliable and safe operation of a smart grid by providing critical operation information. Traditional load forecasting studies mainly deal with the uncertainties of load demand. Different from the traditional load demand, the net load is a compilation of different uncertain quantities such as load, wind, and solar generation to a single uncertain quantity [4]. It actually refers to the effective load in a system and is estimated by taking the difference between total load and renewable generation. In this case, accurate net load forecasting is essential and can benefit different applications with different prediction horizons. For example, the hours or days ahead net load information can be used to support system operations such as adjusting the regional balance between load demand and scheduled supply and the reserve margins, optimally scheduling dispatchable generation units, and estimating power system flexibility requirements [5]. Closer short-term net load forecasting, such as 5 min–30 min ahead, can benefit the reliable power supply or real-time economic dispatch frame in power markets. Therefore, the short-term net load prediction problem is one of the significant problems in renewable energy systems. However, in the case of the aggregation of the individual wind and solar, the net load is much more difficult to predict due to the intermittent nature of the meteorological conditions and its stochastic variability, which prompts the forecasting models to tackle more uncertainties and larger forecasting errors [6]. The stochastic uncertainty [7] for load forecasting tasks has been widely investigated. Different regression

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techniques are used to capture the stochastic uncertainty of net load pattern, such as fully connected network (FCN) [8], long-short term memory (LSTM) [9,10], convolutional neural network (CNN) [11,12], quantile regression (QR)-based method [13,14]. With the help of a large number of historical data from consumers, these methods can capture the stochastic uncertainty of net load patterns and establish the mapping from the collected information to future net loads.

However, the construction of a prediction model may be different for varying load and renewable energy conditions. For the specific prediction methods, Ref. [15] indicated that a universally best method simply does not exist. The selection of forecasting algorithms and the construction of parametric prediction models are critical for the prediction results. For example, the final performance may be affected by the settings of hyperparameters, data preparation, and feature engineering. In addition, pure parametric models also suffer more contingencies in the training process, such as parameter initialization and random gradient updates. These factors that affect the results of the prediction model can be summarized as epistemic uncertainty [7]. In the actual use of the parametric prediction models, epistemic uncertainty is an inevitable problem that the operators have to face in more difficult net load tasks. It makes the final training results of the models face more forecasting errors and dissatisfying prediction intervals.

Under this background, it is of great importance to explore an effective method to deal with the multiple uncertainties and achieve more accurate probabilistic prediction for net load, which is the main motivation of this paper. Gaussian process regression (GPR) is a nonparametric method based on Bayesian theory. It has advantages over parametric methods, especially in regression analysis of a small amount of training samples. In this case, GPR can be applied to model the forecasting residual through the lasted net load pattern and revise the outcome of the original forecasting model. This process improves the accuracy of the original model and copes with the epistemic uncertainty. In addition, GPR allows us to quantify the stochastic uncertainty of the net load through prediction interval, which is very helpful for the planning and operation of the safety system. Note that the proposed residual learning method can be implemented for any deterministic prediction model to deal with the interference of uncertainties. The main contributions of this article are summarized as follows.

1) This paper proposes a probabilistic net load forecasting method based on a Transformer network and Gaussian process-enabled residual modeling learning (RML) method. The Transformer network is first applied to discover the complex structure in net load utilizing abundant historical data. Then, the forecasting errors are modeled by a Gaussian process (GP) to capture the prediction behavior of the pre-trained Transformer network. The adopted composite kernel allows the proposed method to incorporate the advantages of the neural network as well as Gaussian process methods and thus achieve accurate probabilistic forecasting results.
2) This paper designs a Transformer structure-based net load forecasting model. The model can adaptively fuse different types of features and select important pattern information, which benefits from the utilization of cross- and self-attention machines. Different from the recurrent neural network, the Transformer structure have learning capabilities for longer range dependence and use attention to evaluate their importance, which enables it to have better robustness to sudden power changes and events.
3) The proposed RML method can be implemented behind any deterministic prediction model to quantify the uncertainties of point forecasting errors according to the latest load pattern, which can upgrade existing point forecasting models by improving their forecasting performance and converting them into probabilistic predictions. This distinguishes the proposed RML method from existing probabilistic net load forecasting methods that have higher epistemic uncertainty.

The rest of this article is organized as follows. Section II provides a literature review of load forecasting methods. Section III describes the proposed Transformer network and RML method. Section IV compares and analyzes the forecasting results of varying methods. Finally, this article is concluded in Section V.

2. Related works

Traditional load forecasting methods can be classified into two categories: the point forecasting method and the probabilistic method. In Ref. [16], various forecasting methods have been summarized. Ref. [17] proposed a multi-variable linear regression (LR)-based short-term load forecasting model. Multi-variable LR methods may suffer from providing high-accuracy forecasting results when facing complex nonlinear systems. Ref. [18] proposed an intraday load prediction model based on the autoregressive integrated moving average (ARIMA) method. The original load is regarded as the random variable of the model, and the changing process of these original loads is deduced according to the law of statistics [19]. In Ref. [20], Support vector machine Regression (SVR) based methods were constructed for short-term load forecasting. In recent years, neural network-based methods have been widely used for load forecasting in literature owing to their strong non-linear fitting capability. Ref. [8] proposed an FCN-based load forecasting model and obtained the forecast interval with copula theory. Ref. [21] employed deep belief networks (DBN)-based models to perform the short-term load forecasting using restricted Boltzmann machine (RBM) as feature extractors. Recently, various recurrent neural network (RNN)-based methods have been proposed for load forecasting, such as LSTM [9] and gated recurrent units (GRU) [10] methods. Ref. [9] proposed an LSTM-based structure to capture the uncertainties of the net load. In addition, CNN [11], and CNN-LSTM [12] have also been implemented in load forecasting and achieved satisfactory results.

These methods can achieve point forecasting, which can provide a single value for each step. Owing to the increase in renewable energy penetration and market competition, probabilistic forecasting methods are increasingly required by the system operator. The probabilistic forecasting method can quantify uncertainties through intervals, quantiles, or probability density functions (PDF), which can be utilized for unit commitment, reliability planning, and bidding in the power market [22]. Different probabilistic forecasting methods have been proposed, which can be divided into two groups [23]: one-step methods [13,14] and two-step methods [24,25]. Ref. [26] proposed a GP-based method for electricity demand forecasting. The Bayesian characteristics of the GP-based methods can be used to quantify the uncertainty of the load variation [27]. Ref. [28] proposed quantile-based methods based on the LSTM models, which can perform multi-step ahead forecasting and estimate the prediction intervals at any resolution. Ref. [29] introduced the variational autoencoder to perform data generation and build the prediction interval through the sampling process.

The forecasting task of the net load refers to the difference between power consumption and energy generation. In addition, net load forecasting is more difficult than traditional load forecasting since the uncertainty of renewable energy generation should also be considered. In the past few years, several studies have focused on the forecasting of net load [30–34]. Ref. [30] proposed a point forecasting method for PV and load based on graph neural network (GNN) methods, which can utilize the spatial correlation of the features. Ref. [31] compared the performance of the integrated net load forecasting model and the additive model. Ref. [32] proposed a net load forecasting method, which modeled the predictive distributions with a Generalized Pareto distribution. Ref. [34] proposed a deep quantile regression method to achieve net-load forecasting at the distribution level. Ref. [7] proposed a probabilistic prediction model based on the Bayesian theory and LSTM. Simulation results illustrated a superior performance utilizing both deterministic and probabilistic metrics. These research results can deal with multiple uncertainties through special model structures and more

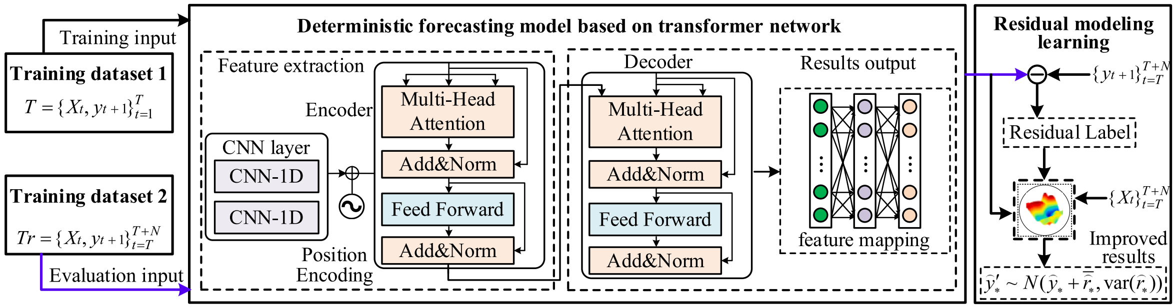


Fig. 1. Proposed Transformer-structure network and residual modeling learning-based probabilistic net load forecasting method.

complex learning methods. However, it may be difficult for operators to construct and adjust the forecasting models in practice. Instead of enhancing the original forecasting model with a more complex process, the proposed method solves these uncertainties by learning the knowledge of forecasting residuals. Through the RML method, the proposed method can improve the performance of the original model and quantify the uncertainties of the forecasting results.

3. Proposed residual modeling learning

In this section, the probabilistic net load forecasting problem is first formulated, and a detailed description of the Transformer network and the RML method based on GPR are illustrated. Finally, the details of the training procedure are provided. The schematic of the proposed Transformer-RML is shown in Fig. 1. The Transformer network is employed to construct the mapping from features to future net load and the RML method is utilized to revise the deterministic outcomes and to quantify the forecasting uncertainty. The detailed descriptions of the algorithms are shown in the following.

3.1. Formulation of the probabilistic load forecasting problem

Consider the net load data recorded by smart meters as a time series $\{P_{i,t}\}$. To estimate future $P_{i,t}$, historical net load data, and other variables such as meteorological data and time index are acquired to build the dataset for training. In specific, the dataset is composed of output variables $yt+1$, which represent the future net load to predict, and input variables $\boldsymbol{\theta t}$ which are a combination of variables that are related to $yt+1$. Net load forecasting aims to learn the mapping relationship $\hat{y}t+1=g(\boldsymbol{\theta t})$ according to the training dataset $T=\{\boldsymbol{\theta t}, yt+1\}_{t=1}^{T}$, where T is the number of instances in the training dataset. Directly estimating the function $g(\cdot)$ can provide deterministic forecasting results of net load. However, the actual values of net load contain uncertainties in both the generation side and the demand side, the point forecasting results may suffer from large prediction errors. The better construction and parameters of forecasting models are also difficult to find directly. In this case, this article proposes a forecasting framework based on NN and Bayesian theory that estimates the distribution of function $g(\cdot)$, which can better capture the uncertainties of net load and provide probabilistic forecasting results. Consider a relatively abundant training dataset as $T=\{\boldsymbol{\theta t}, yt+1\}_{t=1}^{T}$ and the latest dataset with a small amount of data as $Tr=\{\boldsymbol{\theta t}, yt+1\}_{t=T}^{T+N}$ and a test dataset as $Ts=\{\boldsymbol{\theta t}, yt+1\}_{t=T+N}^{S}$. The Transformer network will construct the original deterministic model with the help of the dataset T and the revision process will be achieved with a dataset Tr based on the RML kernel. The detailed illustrations of the proposed method are given below.

3.2. Transformer model and cross-attention mechanism

The Transformer utilizes the seq2seq framework. Generally, a Transformer module mainly consists of self-attention and feed forward networks (FFN), which is the basic cell of the Transformer network. In this research, the Transformer-based encoder and decoder are utilized to construct the original deterministic net load forecasting models. When processing temporal input information, the self- and cross-attention mechanisms are employed to select critical inputs and thus increase the performance of the neural networks. The $\boldsymbol{\theta t}$ in training dataset $T=\{\boldsymbol{\theta t}, yt+1\}_{t=1}^{T}$ can be considered as $\boldsymbol{\theta t}=\{\boldsymbol{Ht}, \boldsymbol{St}\}$, where $\boldsymbol{Ht}\in\mathbb{R}^{L1\times dmodel}$ denotes the past net load data close to $yt+1$ and $\boldsymbol{St}\in\mathbb{R}^{L2\times dmodel}$ denotes other available components related to $yt+1$. Here, $L1$ and $L2$ mean the length of the temporal input information for $\boldsymbol{Ht}$ and $\boldsymbol{St}$, respectively. Their feature dimension at every time point is represented as $dmodel$. The net load forecasting models establish the mapping $\boldsymbol{\theta t}\rightarrow yt+1$ using the training dataset $T=\{\boldsymbol{\theta t}, yt+1\}_{t=1}^{T}$. Consider the input load information $\boldsymbol{\theta t}$ in one attention head and the information propagation for $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$ in Transformer module can be calculated by:

$$\boldsymbol{Q}=\boldsymbol{Wi}^{Q}\boldsymbol{S}, \boldsymbol{K}=\boldsymbol{Wi}^{K}\boldsymbol{H}, \boldsymbol{V}=\boldsymbol{Wi}^{V}\boldsymbol{H}, \tag{1}$$

where $\boldsymbol{Wi}^{Q}, \boldsymbol{Wi}^{K}, \boldsymbol{Wi}^{V}\in\mathbb{R}^{dmodel\times d^{k}}$ and $dk=dmodel/h$ represents the learnable parameters for the information mapping from the input embedding, $\boldsymbol{Ht}$ and $\boldsymbol{St}$; h denotes the attention head number; $\boldsymbol{Q}$ represents the query matrix; and $\boldsymbol{K}$ represents the information that requires attention for current tasks. Therefore, the feature embedding $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$ denote the feature information from different perspectives extracted from the input temporal sequence information, respectively. Using these embeddings, the subsequent feature propagation of one attention head can be calculated by:

$$headi=Attention(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}), \tag{2}$$

$$Attention(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V})=\boldsymbol{V}\cdot softmax\left(\frac{\boldsymbol{QK}^{T}}{\sqrt{dk}}\right), \tag{3}$$

where the weight matrix $\boldsymbol{QK}^{T}$ can be considered as the similarity information that calculates the correlation among different time points in the input information. The weights matrix is further employed to extract the key information from $\boldsymbol{V}$ and perform information propagation in one attention head. For stronger memory capacity and feature extraction ability of the Transformer module, multiply attention heads are established to convert input temporal features in parallel. In this case, the output is achieved by fusing the information from all attention heads, which can be formulated by:

$$MultiHead(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V})=Concat(head1, \ldots, headh)\boldsymbol{WO}. \tag{4}$$

Subsequently, a forward propagation is constructed after the attention mechanism to perform information mapping. The calculation of the forward propagation is:

$$FFN(\boldsymbol{x})=max(0, \boldsymbol{x}\boldsymbol{W}1+\boldsymbol{b}1)\boldsymbol{W}2+\boldsymbol{b}2, \tag{5}$$

where $FFN(\boldsymbol{x})$ represents the feed forward network of the feature information; $\boldsymbol{x}$ represents the output matrix from multiple attention heads

$MultiHead(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V})$; and $\boldsymbol{W}1$, $\boldsymbol{W}2$, $\boldsymbol{b}1$, and $\boldsymbol{b}2$ represent the learnable parameters in the forward propagation layer. The multi-head attention layers and FFN layers are stacked through the residual calculation to optimize gradient propagation, which constructs the basic Transformer-encoder/decoder modular [35], seeing Fig. 1 for example. In this article, the Transformer encoder first learns the key pattern characteristics with the close historical net load data $\boldsymbol{H}$ as the input. In this stage, its inner self-attention machine works to analyze the importance and correlation information at series $\boldsymbol{Ht}$ and subsequently generate feasible feature embeddings. After that, the Transformer decoder accepts these embeddings as well as other available components related to the predicted net load $\boldsymbol{S}$ and then adaptively fuses these features through the cross-attention mechanism. Therefore, the generation pattern of the net load and correlation information among every time point in $\boldsymbol{H}$ and $\boldsymbol{S}$ are jointly modeled by the Transformer decoder module. These allow the proposed point forecasting model to capture the complex net load pattern using historical data and be sensitive to abrupt generation changes and events.

3.3. Gaussian process with RML kernel

Gaussian process regression is a non-parametric Bayesian method for dealing with regression problems [36]. It can capture various mapping relationships between input features and outputs utilizing theoretically infinite functions and allows the data to determine the level of complexity through the means of Bayesian inference. In this article, Gaussian process regression is employed to improve the performance of the Transformer network and further quantify the uncertainty of net load. It is achieved by capturing the prediction behavior of the pre-trained Transformer network through residual learning. Specifically, supposing that (X, Y) and (X^*, Y^*) represent the training dataset $Tr = \{Xt, yt+1\}_{t=T}^{T+N}$ and test dataset $Ts = \{Xt, yt+1\}_{t=T+N}^{S}$, where $X \in R^{N\times F}$ and $Y \in R^{N\times 1}$ are the input and output of the training set; $X^* \in R^{N^*\times F}$ and $Y^* \in R^{N^*\times 1}$ are input and output of the test set, where N denotes the sample number and F represents the feature number. N^* is the volume of the test dataset. Then we can obtain

$$Y = f(X) + \varepsilon \tag{6}$$

$$Y^* = f(X^*) + \varepsilon \tag{7}$$

where $\varepsilon \sim N(0, \sigma_n^2)$ denotes Gaussian white noise; $f(\cdot)$ represents the Gaussian process mapping from the input to output:

$$f(x) = GP(m(x), k(x, x')) \tag{8}$$

where $x, x' \in X$; $m(\cdot)$ is the mean function; $k(\cdot)$ denotes the covariance function, among which the radial basis function (RBF) is the most commonly used one:

$$k(x, x') = \sigma^2\ exp\ (-(x - x')^2 / 2) \tag{9}$$

Then we can obtain the prior Gaussian distribution of y:

$$y \sim N\left(0, k(X, X), \sigma_n^2 I_n\right) \tag{10}$$

and the joint distribution of observed value y and predicted value f_* can be obtained as:

$$\begin{bmatrix} y \\ f_* \end{bmatrix} \sim N\left(0, \begin{bmatrix} k(X,X) + \sigma_n^2 I_n & k(X, x^*) \\ k(x^*, X) & k(x^*, x^*) \end{bmatrix}\right) \tag{11}$$

Then we can obtain the posterior distribution of the predicted value

$$f_*|X, y, x_* \sim N(\overline{f}_*, \mathrm{cov}(f_*)), \tag{12}$$

$$f^* = k(x^*, X)\left(k(X, X) + \sigma_n^2 I_n\right)^{-1} y, \tag{13}$$

$$\mathrm{cov}(f_*) = k(x^*, x^*) - k(x^*, X) \times \left(k(X, X) + \sigma_n^2 I_n\right)^{-1} k(X, x^*). \tag{14}$$

Different from typical GP that models the mapping between available input X and power output Y, RML aims to model the residuals between the labels y and the prediction value $\widehat{y}$ calculated by the Transformer utilizing GP with a synthetic kernel [37]. The residuals between the actual power values and the prediction net load values of the training set are

$$r_i = y_i - \widehat{y}_i, i = 1, 2, ..., n. \tag{15}$$

$\widehat{Y}$ and R represent the net load outcomes calculated by the Transformer network and all residuals, respectively. A Gaussian process with a synthetic kernel is trained to assume $R \sim N(0, k_c((X, \widehat{Y}), (X, \widehat{Y})) + \sigma_n^2 I)$, where $k_c((X, \widehat{Y}), (X, \widehat{Y}))$ represents the covariance matrix of all data points in the training set based on the following synthetic kernel:

$$k_c\left(\left(x_i, \widehat{y}_i\right), \left(x_j, \widehat{y}_j\right)\right) = k_{in}(x_i, x_j) + k_{out}\left(\widehat{y}_i, \widehat{y}_j\right), i = 1, 2, ..., n. \tag{16}$$

Supposing that the Radial basis function (RBF) kernel is utilized for both $k_{in}(\cdot)$ and $k_{out}(\cdot)$. Then the synthetic kernel can be denoted as

$$k_c\left(\left(x_i, \widehat{y}_i\right), \left(x_j, \widehat{y}_j\right)\right) = \sigma_{in}^2\ exp(- \|x_i - x_j\|^2 / 2l_{in}^2) + \sigma_{out}^2\ exp\left(-\left\|\widehat{y}_i - \widehat{y}_j\right\|^2 \Big/ 2l_{out}^2\right), \tag{17}$$

where σ_{in}, σ_{out}, l_{in}, and l_{out} represents the parameters to be optimized during the learning process, which are calculated by maximizing the function of log marginal likelihood as:

$$\log p\left(R\middle|X, \widehat{Y}\right) = -\frac{1}{2}R^T\left(k_c\left(\left(X, \widehat{Y}\right), \left(X, \widehat{Y}\right)\right) + \sigma_n^2 I\right)^{-1} R - \frac{1}{2}\log\left|k_c\left(\left(X, \widehat{Y}\right), \left(X, \widehat{Y}\right)\right) + \sigma_n^2 I\right| \frac{n}{2}\log 2\pi. \tag{18}$$

When the learning process is finished, the distribution function of the prediction residual and a clean input x_* can be calculated as $\widehat{r}_*\Big|X, \widehat{Y}, R, x_*, \widehat{y}_* \sim N(\overline{\widehat{r}}_*, \mathrm{var}(\widehat{r}_*))$ by the trained Gaussian process, where

$$\overline{\widehat{r}}_* = k_*^T\left(k_c\left(\left(X, \widehat{Y}\right), \left(X, \widehat{Y}\right)\right) + \sigma_n^2 I\right)^{-1} R, \tag{19}$$

$$\mathrm{var}\left(\widehat{r}_*\right) = k_c\left(\left(x_*, \widehat{y}_*\right), \left(x_*, \widehat{y}_*\right)\right) - k_*^T\left(k_c\left(\left(X, \widehat{Y}\right), \left(X, \widehat{Y}\right)\right) + \sigma_n^2 I\right)^{-1} k_* \tag{20}$$

Then the calculated residual is utilized to revise the point forecasting result of the Transformer network. The revised probabilistic forecasting result based on Gaussian distribution is shown as:

$$\widehat{y}'_* \sim N\left(\widehat{y}_* + \overline{\widehat{r}}_*, \mathrm{var}\left(\widehat{r}_*\right)\right). \tag{21}$$

Two features distinguish the proposed Transformer-RML probabilistic forecasting framework from the previous studies. Firstly, the proposed method fully utilizes the strong temporal feature extraction capability of the Transformer network, which mitigates the shortcomings of the GP in dealing with time series data and more difficult forecasting tasks. Secondly, in addition to providing uncertainty quantification ability, the utilization of the RML method can also improve the performance of the point net load forecasting method through Bayesian characteristics. Therefore, the superiorities of the proposed Transformer-RML probabilistic method in real applications are as follows. 1) The proposed method can achieve high-performance probabilistic forecasting in dealing with net-load uncertainty from renewable energy, which contributes to reliable prior net load information for system operations. 2) The proposed RML method can convert the point forecasting results to probabilistic information, which allows it to update the existing point net load forecasting models without retraining and redesigning processes. 3) The proposed RML method can improve the performance of point forecasting models, which mitigates the epistemic uncertainty from neural networks facing net load tasks. 4) The RML method has less training cost because it is trained with just a few days of data, which allows it to be deployed quickly in reality. The detailed training process and deployment of the proposed method are shown as follows.

Algorithm Training process and deployment of the proposed Transformer-RML method

Model training of the Transformer network and GP-based RML methods
Input: Training dataset $T = \{\theta t, yt+1\}_{t=1}^{T}$ and the latest dataset with a small amount of data $Tr = \{\theta t, yt+1\}_{t=T}^{T+N}$
Output: point forecasting Transformer model $g(\cdot)$, RML-based probabilistic forecasting model $f(\cdot)$
1: Initialize the parameters of the Transformer network $g(\cdot)$
2: **for** epoch $t = 1, 2, \ldots, P$ **do**
3: Obtain a stochastic batch of data from the training dataset $T = \{\theta t, yt+1\}_{t=1}^{T}$
4: Calculate the gradient by minimizing the L1 loss function
5: Update the parameters of $g(\cdot)$ according to the gradient rule
6: **end**
7: Obtain the point forecasting Transformer model $g(\cdot)$
8: Load the parameters of $g(\cdot)$ and calculate the outcomes on the dataset $Tr = \{\theta t, yt+1\}_{t=T}^{T+N}$
9: Obtain the vector of residuals $r_i = y_i - \widehat{y}_i, i = 1, 2, \ldots, N$
10: Stack the vector X_*, $\widehat{Y}_*$, and R
11: Initialize the parameters of GP and obtain the prior model $R \sim N(0, K_c((X, \widehat{Y}), (X, \widehat{Y})) + \sigma_n^2 I)$
12: Optimize the parameters by maximizing $\log p(R|X, \widehat{Y})$
13: Obtain the posterior model $f(\cdot)$ for the residual prediction
Model deployment of proposed probabilistic forecasting method
Input: Test dataset $Ts = \{\theta t, yt+1\}_{t=T+N}^{S}$
Output: probabilistic forecasting outcomes for $Ts = \{\theta t, yt+1\}_{t=T+N}^{S}$
1: Calculate the point forecasting outcomes through $\widehat{y}_* = g(\theta t)$
2: Calculate the revised residuals through $\overline{\widehat{r}}_* = f(\theta t, g(\theta t))$
3: Obtained revised point forecasting outcomes $\widehat{y}_* + \overline{\widehat{r}}_*$
4: Quantify the uncertainty of the revised outcomes through $\widehat{y}'_* \sim N(\widehat{y}_* + \overline{\widehat{r}}_*), \mathrm{var}(\overline{\widehat{r}}_*))$

4. Case study

4.1. Data description

Numerical comparative tests are carried out using real-field smart meter data. The data are measurement data of a distribution network in Ausgrid, containing the load demand and PV generation data of 300 households in residential areas with accurate PV smart meters operating from July 1, 2011, to June 30, 2012 [38]. The PV generation and the load consumption of 300 households are summed together to simulate the net load demand of a microgrid node in the power grids. The sampling interval of the recorded data is 30 min. For more details about the dataset, readers can refer to Ref. [38]. This study aims to achieve point and probabilistic forecasting of the aggregated net load. Accurate net load forecasting can benefit different applications with different prediction horizons. In this article, two kinds of one-step forecasting tasks are implemented to verify the effectiveness of the proposed method, including predicting the net load values in 30 min and 12 h, respectively.

Five dataset position cases are considered to simulate various seasons, including: Case 1) the training dataset at 0th-100th days and the testing dataset at 100th-110th; Case 2) the training dataset at 50th-150th days and the testing dataset at 150th-160th; Case 3) the training dataset at 100th-200th days and the testing dataset at 200th-210th; Case 4) the training dataset at 150th-250th days and the testing dataset at 250th-260th; Case 5) the training dataset at 200th-300th days and the testing dataset at 300th-310th.

The selected features include the following variables: 1) The time index, which contains the hour index in the day, the day index in the week, and the month index in the year. In this study, one-hot encoding is used to encode the time index; 2) The past historical net load data, which contains the load data at the same time-step of the past 48 time points; 3) For the proposed method, the data close to the prediction time is taken as feature $\boldsymbol{H}$ (in past 24 time points) and the data relatively far from the prediction time is taken as feature $\boldsymbol{S}$ (out of past 24 time points). Note that the forecasting tasks in this article only use the aggregated net load data, and the PV data from each customer are invisible.

4.2. Evaluation metrics

To evaluate the performance of the point forecasting results of various methods, three widely used deterministic forecasting evaluation metrics and two probabilistic forecasting metrics are utilized in this study.

1) *Metrics for deterministic forecasting*

The point forecasting metrics are mean absolute error (MAE) and root mean square error (RMSE):

$$MAE = \frac{1}{m} \sum_{t=1}^{m} \left| \widehat{y}_t - y_t \right| \tag{22}$$

$$RMSE = \sqrt{\frac{1}{m} \sum_{t=1}^{m} \left(\widehat{y}_t - y_t \right)^2} \tag{23}$$

where y_t and $\widehat{y}_t$ are the actual values and predicted values of the *t-th* sample; m is the sample number in the test dataset. MAE calculates the average absolute value between the actual load demand. RMSE tends to penalize the situations when there exist large deviation values.

2) *Metrics for probabilistic forecasting*

Pinball loss and Winkler probabilistic forecasting metrics are used in this study. Pinball is a comprehensive indicator that can evaluate the above metrics, which is defined as:

$$Pinball = \begin{cases} (1-q)\left(\widehat{y}_{t,q} - y_t\right), y_t < \widehat{y}_{t,q} \\ q\left(y_t - \widehat{y}_{t,q}\right), y_t \geq \widehat{y}_{t,q} \end{cases}, \tag{24}$$

where q ($q = 0.01, 0.02, \ldots, 0.99$) represents the quantiles and $\widehat{y}_{i,q}$ denotes the forecasting outcomes at the q-th quantile. A lower Pinball value indicates a better prediction performance. The Winkler loss is calculated by:

$$Winkler = \begin{cases} \left(U_{t,\alpha} - L_{t,\alpha}\right) + 2\left(L_{t,\alpha} - y_t\right)/\alpha \text{ if } y_t < L_{t,\alpha} \\ \left(U_{t,\alpha} - L_{t,\alpha}\right) \quad \text{if } L_{t,\alpha} \leq y_t \leq U_{t,\alpha} \\ \left(U_{t,\alpha} - L_{t,\alpha}\right) + 2\left(y_t - U_{t,\alpha}\right)/\alpha \text{ if } y_t > U_{t,\alpha} \end{cases}, \tag{25}$$

where $U_{t,\alpha}$ and $L_{t,\alpha}$ denote the upper and lower bounds of $100(1-\alpha)\%$ forecasting interval at the time point t respectively. The following is used to calculate the prediction interval coverage probability (PICP):

$$PICP = \frac{1}{m} \sum_{t=1}^{m} ct, ct = \begin{cases} 1, if \ \ yt \in \left[L_{t,\alpha}, U_{t,\alpha}\right] \\ 0, if \ \ yt \notin \left[L_{t,\alpha}, U_{t,\alpha}\right] \end{cases}. \tag{26}$$

The prediction interval average width (PIAW) is used to calculate the coverage of the actual values using the prediction intervals. The following is employed to calculate the PIAW values:

Table 1
Hyperparameter selection of the proposed Transformer network.

Metrics	Number of attention heads			
	2	4	8	16
MAE (kW)	9.00	**8.64**	9.14	9.22
RMSE (kW)	11.8	**11.5**	11.9	12.3
Metrics	Ratio of dropout			
	0.00	0.05	0.1	0.2
MAE (kW)	9.58	**8.64**	9.80	10.3
RMSE (kW)	12.2	**11.5**	12.9	13.7
Metrics	Dimension of hidden layers			
	32	64	128	256
MAE (kW)	10.2	9.69	**8.64**	9.09
RMSE (kW)	13.5	12.7	**11.5**	11.9

Table 2
Structure of the proposed Transformer network.

Layer	Layer hyperparameters (scanning field, number of filters)	
Input layer	Past net load data ***Ht***	Future available data ***St***
1	CNN1d (3,128)	CNN1d (3, 128)
2	Max pooling (2)	Max pooling (2)
3	CNN1d (3, 128)	CNN1d (3, 128)
4	Position encoding	Position encoding
5	Transformer-Encoder	–
6	Obtaining ***K***, ***V***	Obtaining ***Q***
7	Transformer-Decoder	
8	Dropout (0.05)	
9	Fully connected layer (1, 128)	
10	Fully connected layer (1, 1)	
Output layer	$\hat{y}t+1$	

$$PIAW_{\alpha} = \frac{1}{m}\sum_{t=1}^{m}\left(U_{t,\alpha} - L_{t,\alpha}\right), \tag{27}$$

where $PIAW_{\alpha}$ is the $PIAW$ for $100(1-\alpha)\%$ prediction interval. The PICP and PIAW are sets of corresponding indicators. The estimation intervals should satisfy high PICP and low PIAW. The continuous ranked probability score (CRPS) is also utilized to evaluate the performance of the various methods, which is calculated by:

$$CRPS(F, yt) = E_F\left|\,\hat{y}_t - yt\right| - \frac{1}{2}E_F\left|\,\hat{y}_t - \hat{y}_t^{'}\right|, \tag{28}$$

where $\hat{y}_t$ and $\hat{y}_t^{'}$ are two independent random variables from the predicted distribution function F at t time point. To eliminate the influence of sequence length, the calculated CRPS, Pinball, and Winkler loss for each sample in the testing set will be added and divided by the sequence length m.

4.3. Experimental setup

The grid search method is employed to determine the hyperparameter of the proposed Transformer network. The three important hyperparameters from the Transformer network are adjusted gradually to find the feasible model architecture for forecasting tasks. The task is performed on a one-step forecasting task to predict the net load in 30 min.

The prediction performance of various hyperparameters is shown in Table 1, where the best results are bold. It can be observed from the table that the number of attention heads, the ratio of dropout, and the dimension of the hidden layer should be set as 4, 0.05, and 128, respectively. Therefore, the original deterministic Transformer model designed in this paper is shown in Table 2. Here, CNN1d means the one-dimensional convolutional neural network.

The comparative methods are separated into two groups: the point forecasting methods and the probabilistic methods. The comparative

Table 3
Hyperparameter sets of various methods.

Model	Hyperparameter settings
SVR	kernel = RBF
SGP	Number of inducing point = 100, kernel = RBF
GBR/QGBR	Number of estimators = 20, max_depth = 3, learn_rate = 0.1
CNN	CNN structure = [128, 128], FCLs structure = [128, 1], activation function = ReLU, learn_rate = 2 × 10e^{-4}
LSTM/ QLSTM	LSTM structure = [128, 128], FCLs structure = [128, 1/99], activation function = ReLU, learn_rate = 2 × 10e^{-4}
ENN	FCLs structure = [128, 128, 1], activation function = ReLU, num_clusters = 20, learn_rate = 2 × 10e^{-4}
DropoutNN	FCLs structure = [128, 128, 128, 1], activation function = ReLU, Dropout = 0.2, learn_rate = 2 × 10e^{-4}, sampling number = 500
VENN	FCLs structure = [128, 128, 128, 1], activation function = ReLU, learn_rate = 2 × 10e-4, sampling number = 500
RML	Number of inducing point = 100, kernel = RBF + RBF

point forecasting methods include: 1) convolutional neural network (CNN) [11] method, where CNN1d layers and fully connected layers (FCLs) are employed to capture the regression rules; 2) long short-term memory network (LSTM) [9] method, where LSTM layers and FCLs are utilized; 3) Transformer network which has the same structure as the proposed method; 4) gradient boosting regression (GBR) [39] method, where numerous weak learners are aggregated into an ensemble; 4) support vector regression (SVR) [20] method, where RBF is selected as the kernel function; 5) persistence method (PM), where the latest actual values or the actual values at the same time in the past day are directly considered as the predicted values.

For the comparative probabilistic forecast methods, six cases are investigated, including: 1) quantile gradient boosting regression (QGBR) method, where each QGBR model is developed to predict the quantile of one level, resulting in a total of 99 QGBRs for interval predictions; 2) sparse Gaussian process (SGP) [40] method, where sparse technique is used to accelerate the training process of GP, which reduce the computational complexity from $O(e^3)$ of original GP to $O(c^2e)$. Here, e means the number of training samples and c means the number of inducing points; 3) quantile long short-term memory network (QLSTM) [41], which generates predictions of 99 quantile values; 4) ensemble neural network (ENN) [42] method, where the outputs of numerous independent neural networks are modeled through Gaussian distribution; 5) dropout neural network (DropoutNN) [43] method, where the outputs of a neural network with dropout layers are modeled through Gaussian distribution; 6) variational encoder neural network (VENN) [44] method, where a variational encoder is employed for generation through sampling, followed by a neural network to output the predicted values and its outputs are modeled through Gaussian distribution. For fairness in comparison, the hidden layer dimensions in neural network-based comparative methods are set as 128. The hyperparameter sets of various methods are shown in Table 3. The settings of the GBR method, learning rate, and layer number for neural network-based methods can be referred to Ref. [28]. For the SVR, QGBR, and SGP methods, all training datasets are used to train the models. For the comparative neural network-based methods, the last 10 days at the training dataset are used as a validation set to select the model parameters. For the proposed method, the last 5 days at the training dataset are used for RML on the basis of the Transformer network.

4.4. Point forecasting results

Comparative tests are carried out in predicting the net load in 30 min to verify the effectiveness of the proposed forecasting method. Five dataset cases are considered to simulate various seasons. The comparative methods include: 1) CNN; 2) LSTM; 3) Transformer network; 4) GBR; 4) SVR; 5) PM, where the latest actual values are directly considered as the predicted values.

The MAE and RMSE achieved by various forecasting methods at 30

Table 4
MAE and RMSE achieved by various forecasting methods at 30 min forecasting task.

Metrics	Models	Training dataset positions				
		Case 1	Case 2	Case 3	Case 4	Case 5
MAE (kW)	Proposed	**7.34**	**7.22**	8.17	**6.37**	**7.64**
	CNN	8.24	7.55	**7.81**	7.45	9.54
	LSTM	11.3	8.49	11.3	8.20	10.9
	Transformer	9.66	7.80	8.59	6.49	8.64
	GBR	11.4	9.32	8.91	8.93	11.2
	SVR	18.3	10.2	12.9	11.3	14.7
	PM	13.1	10.9	9.84	11.7	13.6
RMSE (kW)	Proposed	**9.92**	**9.56**	11.5	**8.58**	**10.3**
	CNN	10.5	9.65	**10.9**	10.6	13.2
	LSTM	15.4	11.3	17.1	12.8	16.3
	Transformer	12.8	10.4	12.1	8.73	11.5
	GBR	15.1	12.5	14.8	11.7	15.1
	SVR	24.6	13.3	22.0	14.6	18.7
	PM	17.0	14.2	12.9	15.2	16.9

Table
5MAE and RMSE achieved by various forecasting methods at 12 h forecasting task.

Metrics	Models	Training dataset positions				
		Case 1	Case 2	Case 3	Case 4	Case 5
MAE (kW)	Proposed	**16.0**	**18.4**	**19.0**	**18.6**	**15.4**
	CNN	19.4	21.8	19.7	20.6	16.9
	LSTM	18.8	21.3	19.7	19.9	16.5
	Transformer	17.6	20.0	19.2	18.7	16.0
	GBR	19.6	21.4	19.4	18.9	17.1
	SVR	22.9	21.6	21.4	21.5	18.6
	PM	20.1	26.0	20.5	26.1	21.6
RMSE (kW)	Proposed	**26.1**	**27.7**	28.7	**29.5**	**21.4**
	CNN	29.7	35.4	30.5	33.8	25.2
	LSTM	27.7	31.9	28.9	32.9	22.9
	Transformer	27.0	29.3	29.2	30.7	22.7
	GBR	32.5	33.4	**28.5**	32.8	24.2
	SVR	32.7	30.8	30.9	32.7	24.5
	PM	29.6	37.8	30.8	42.4	29.3

min forecasting task are listed in Table 4, where the best results are bold. It can be observed from the table that the SVR method can hardly capture the pattern of load variation and its forecasting outcomes are even worse than the PM, seeing their MAE values under Case 1 and Case 5. By contrast, the LSTM and Transformer methods can derive more correlation information from time sequences, which enables them to outperform traditional GBR and SVR methods in all dataset cases. Different from the pure neural network methods, the proposed method employs a GP to model the residuals, which further improves the forecasting performance from the basis of the Transformer network. In this case, the proposed method can effectively capture the pattern of load variation and thus outperform other methods under most dataset cases. For example, the proposed method can achieve 24% performance improvement compared with the Transformer method at MAE performance under Case 1. The results demonstrate the superiority of the

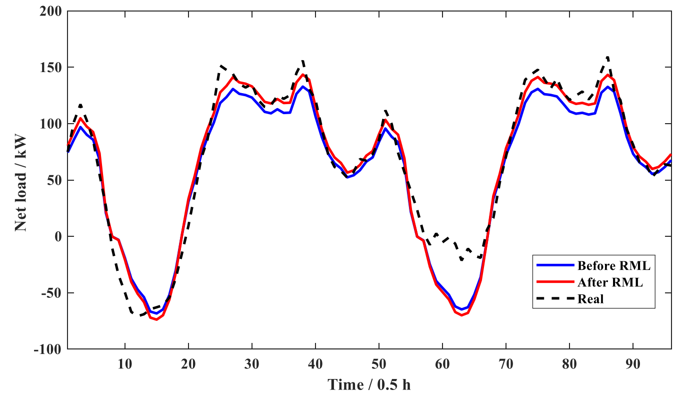


Fig. 3. Revision effect for original Transformer network based on RML method.

proposed residual modeling-based forecasting methods under different point forecasting tasks. More tests are carried out to predict the net load values in 12 h. Five dataset cases are considered to simulate various seasons. For the PM, the actual values at the same time in the past day are directly considered as the predicted values. The settings of other comparative models are the same as 30 min forecasting tasks.

The MAE and RMSE achieved by various forecasting methods at 12 h forecasting task are listed in Table 5, where the best results are bold. It can be seen from the table that all methods suffer larger forecasting errors compared with predicting net load in 30 min due to the increase in task difficulty. Among them, the SVR method can hardly capture the pattern of load variation and its forecasting outcomes are even worse than the PM. By contrast, the LSTM and Transformer methods can model the correlation of sequence information, which enables them to achieve better forecasting performance compared with traditional GBR and SVR methods. Based on the results of the Transformer method, the proposed method employs a GP to model the residuals from the latest dataset, which enables the proposed method to capture the pattern of load variation and thus provides better point forecasting performance under most dataset cases. For example, the proposed method can achieve an 8% performance improvement compared with the original Transformer model at MAE performance under Case 2. The results demonstrate the superiority of the proposed residual modeling-based forecasting method.

To further analyze the performance of the proposed model, the forecasting results of different methods in the test dataset for 3rd May to 6th May are shown in Fig. 2. It can be seen from the figure that the proposed method is closest to the actual load value and has obvious advantages in predicting curve valleys, seeing its predicted results at t = 5–10, 50–55, and 94–105 for example. For the net load, the curve valley represents the peak of PV generation. When t = 50, there is an unconventional PV output variation, which makes the net load fluctuate differently. The other methods cannot detect this variation due to the scarce pattern appearing in the training dataset. By contrast, the proposed Transformer method can reflect this variation. Similar phenomena can be seen at t = 100. The PV generation is larger than normal

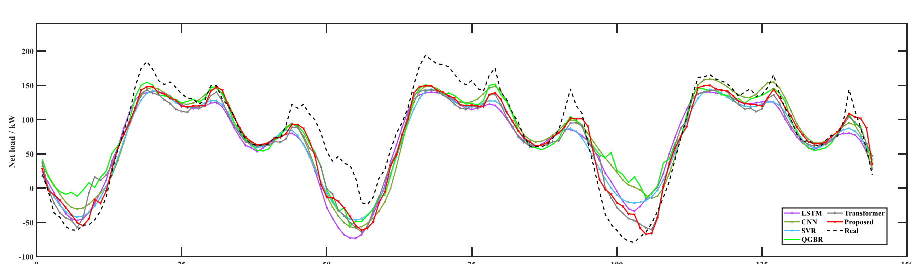


Fig. 2. Deterministic forecast results of different methods.

Table 6
CRPS, Pinball loss, and Winkler score (α = 0.1) obtained by various methods at 30 min forecasting task.

Metrics	Models	Training dataset positions				
		Case 1	Case 2	Case 3	Case 4	Case 5
CRPS (kW)	Proposed	**5.517**	**5.350**	6.164	**4.727**	**5.690**
	QGBR	11.40	7.126	8.207	7.817	9.543
	SGP	6.466	6.549	6.201	6.247	6.836
	QLSTM	11.97	9.044	8.715	7.092	7.934
	ENN	6.444	6.347	5.740	6.564	8.195
	DropoutNN	8.626	7.169	**5.325**	6.101	7.306
	VENN	10.22	8.871	14.00	10.14	10.09
Pinball (kW)	Proposed	**2.783**	**2.699**	3.110	**2.387**	**2.874**
	QGBR	5.727	3.561	4.116	3.915	4.790
	SGP	3.259	3.304	3.132	3.152	3.452
	QLSTM	6.050	4.573	4.407	3.591	4.018
	ENN	3.248	3.210	2.893	3.314	4.131
	DropoutNN	4.351	3.619	**2.688**	3.079	3.687
	VENN	5.172	4.490	7.077	5.128	5.101
Winkler (kW) (α = 0.1)	Proposed	**45.24**	**41.99**	**49.71**	**39.36**	**47.90**
	QGBR	154.7	118.5	115.8	116.0	128.2
	SGP	51.32	49.57	57.40	52.24	52.13
	QLSTM	137.5	95.37	99.62	68.22	77.98
	ENN	77.46	62.60	61.47	83.72	116.1
	DropoutNN	112.5	91.61	66.13	78.90	95.40
	VENN	94.09	80.73	154.7	112.7	112.2

conditions so that other methods cannot follow the valley of the net load. The result illustrates that the proposed method can effectively adapt to the randomness of net load with a high proportion of PV access. It is due to that the designed forecasting model has stronger learning capabilities for capturing the patterns of load demand and PV generation.

In addition, the revision process based on the RML kernel in Case 5 is shown in Fig. 3. It can be seen from the figure that the RML-based method can achieve obvious performance improvement for the original forecasting model, see t = 20–40 and 70–80 for example. The original model is slightly inadequate in predicting the peak of the net load. Through the residual learning at the dataset *Tr*, the forecasting model can more effectively adapt to the load demand pattern of peaks. The experimental results illustrate that the designed RML process is valid in solving the epistemic uncertainty of the original forecasting model.

4.5. Probabilistic forecasting results

Comparative tests are carried out among different probabilistic methods to verify the uncertainty quantification capability of the proposed RML model. The task is a one-step forecasting task to predict the net load values in 30 min. Five dataset position cases are considered to simulate various seasons. For the probabilistic forecast methods, six cases are investigated, including 1) QGBR; 2) SGP; 3) QLSTM; 4) ENN; 5) DropoutNN; 6) VENN methods. The CRPS, Pinball loss, and Winkler score (α = 0.1) obtained by various methods at 30 min forecasting task are listed in Table 6, where the best results are bold. It can be seen from the table that the QGBR method faces relatively larger CRPS, Pinball, and Winkler loss under most dataset cases. By contrast, the ENN and DropoutNN methods capture the uncertainty of load variation through ensemble neural networks, which enables it to outperform the QGBR method under different dataset cases. Although the SGP method can achieve feasible uncertainty quantification under different dataset cases, its forecasting errors are still larger than the proposed method in most cases. Different from the quantile-based method, the proposed method directly converts the point forecasting results to probabilistic outcomes through a GP, which is achieved by modeling the residuals on the basis of the Transformer model. In this case, the proposed can incorporate the advantages of neural network and GP methods, which allows the proposed method to outperform the GP method and other neural network-based methods. For example, the proposed method can achieve 32.1% and 27.9% performance improvement compared with SGP and ENN methods at CRPS loss under Case 4. The results demonstrate the superiority of the proposed residual modeling-based probabilistic forecasting methods. More tests are carried out to achieve probabilistic prediction for net load values in 12 h. Five dataset cases are considered to simulate various seasons. The settings of the comparative models are the same as 30 min forecasting tasks.

Table 7
CRPS, Pinball loss, and Winkler score (α = 0.1) obtained by various methods at 12 h forecasting task.

Metrics	Models	Training dataset positions				
		Case 1	Case 2	Case 3	Case 4	Case 5
CRPS (kW)	Proposed	**13.49**	**15.83**	**14.86**	**15.08**	**12.12**
	QGBR	18.76	16.10	15.93	16.09	16.40
	SGP	13.89	17.75	16.71	17.71	13.09
	QLSTM	15.90	16.59	15.30	16.82	13.84
	ENN	16.97	16.72	16.46	15.63	12.80
	DropoutNN	19.43	19.38	16.35	18.65	14.43
	VENN	16.68	20.18	19.61	17.83	14.36
Pinball (kW)	Proposed	**6.790**	**7.969**	**7.503**	**7.609**	**6.106**
	QGBR	9.431	8.099	8.008	8.098	8.253
	SGP	6.987	8.951	8.425	8.927	6.593
	QLSTM	8.044	8.381	7.740	8.510	7.007
	ENN	8.562	8.443	8.303	7.882	6.461
	DropoutNN	9.773	9.733	8.235	9.376	7.226
	VENN	8.427	10.21	9.901	9.003	7.257
Winkler (kW) (α = 0.1)	Proposed	136.8	**132.5**	145.3	154.3	**103.4**
	QGBR	167.9	142.1	160.8	**126.4**	143.3
	SGP	129.6	166.2	146.6	183.4	118.5
	QLSTM	**112.1**	148.2	**128.7**	171.5	120.0
	ENN	164.1	194.2	187.2	204.4	125.5
	DropoutNN	318.2	333.6	253.0	309.9	196.3
	VENN	194.2	209.9	243.9	217.7	138.8

Table 8
CRPS, Pinball, and Winkler loss obtained by various methods at 12 h forecasting task.

Models	CRPS (kW)	Pinball (kW)	Winkler (kW)			
			α = 0.1	α = 0.2	α = 0.3	α = 0.4
Proposed	**12.12**	**6.106**	**103.4**	**84.29**	**71.86**	**62.20**
QGBR	16.40	8.253	143.3	120.7	105.3	91.45
SGP	13.09	6.593	118.5	93.77	78.79	67.99
QLSTM	13.84	7.007	120.0	93.39	78.24	68.51
ENN	12.80	6.461	125.5	91.23	74.81	64.32
DropoutNN	14.43	7.226	196.3	118.4	88.15	71.67
VENN	14.36	7.257	138.8	100.1	82.86	71.75

The CRPS, Pinball, and Winkler loss (α = 0.1) obtained by various methods at 12 h forecasting task are listed in Table 7, where the best results are bold. It can be seen from the table that the QGBR and DropoutNN methods can hardly capture the uncertainty of net load, resulting in relatively large CRPS, Pinball loss, and interval loss under Case 1. By contrast, the ENN method employs numerous neural networks to capture the uncertainty of load variation, which enables it to outperform the DropoutNN and VENN methods under different dataset cases but also increases its computational and memory burden. Although the SGP method can achieve the feasible uncertainty quantification in Case 1 and Case 5, its forecasting performance is insufficient under Cases 2–4. It may be due to that the SGP method lacks effective time series modeling capabilities to cope with more difficult prediction tasks. Different from the SGP method, the proposed method firstly employs a Transformer model to capture the pattern of load variation and then quantify the uncertainty of prediction errors through a GP, which allows it to outperform QLSTM and other methods at CRPS, Pinball, and Winkler loss under the most dataset cases. For example, the proposed method can achieve a 10.9% performance improvement compared with

Table
9PICP achieved by various forecasting methods at 12 h forecasting task.

Models	PICP			
	α = 0.05	α = 0.1	α = 0.2	α = 0.3
Proposed	0.9625	0.9395	0.8895	0.8437
QGBR	0.9541	0.8979	0.8125	0.7354
SGP	0.9500	0.9333	0.8865	0.8375
QLSTM	0.8500	0.7916	0.6812	0.5645
ENN	0.7312	0.6666	0.5729	0.5166
DropoutNN	0.5958	0.5166	0.4312	0.3604
VENN	0.7750	0.6854	0.5666	0.4625

Table
10PIAW achieved by various forecasting methods at 12 h forecasting task.

Models	PIAW (kW)			
	α = 0.05	α = 0.1	α = 0.2	α = 0.3
Proposed	104.8	87.99	68.55	55.44
QGBR	146.3	122.3	99.47	84.56
SGP	109.2	91.69	71.44	57.77
QLSTM	61.43	50.75	38.18	30.23
ENN	67.88	56.96	44.38	35.89
DropoutNN	28.23	23.69	18.46	14.93
VENN	58.23	48.87	38.07	30.79

the SGP method at Pinball loss under Case 2. The results demonstrate the superiority of the proposed residual modeling-based probabilistic forecasting method.

More estimation tests are carried out in Case 5 at 12 h task to evaluate the probabilistic forecasting performance of the proposed method. The CRPS, Pinball, Winkler, PICP, and PIAW at different prediction intervals are calculated and shown in Tables 8–10, respectively. It can be observed from the tables that the QGBR method suffers a relatively large Winkler loss compared with other methods. It is due to that the prediction intervals of the QGBR method are overly wide. Similarly, the prediction intervals of the DropoutNN method are very narrow, resulting in large Winkler loss and lower PICP. When the SGP method is utilized to construct the probabilistic model, its results are still insufficient compared with the proposed method. It may be due to that the feature extraction capability of SGP cannot support the construction of a forecasting model. By contrast, the proposed method first employs a Transformer model to capture the pattern of load variation and then quantifies the uncertainty of prediction errors through a composite kernel to model the residual. The RML method generates the probabilistic prediction on the basis of the Transformer model. This enables the proposed method to combine the advantages of nonparametric and parametric models and quantify the multiple uncertainties of the net load. Therefore, the proposed method can provide larger PICP with narrower prediction intervals, which allows the Pinball performance of the proposed method 7.38% greater than the performance of the SGP method. When α = 0.1, the Winkler of the proposed method is 103.4, which represents a performance gain of 12.7% and 27.8% over the SGP and QGBR methods. The results demonstrate that the prediction results of the proposed RML method have obvious advantages in reliability and accuracy.

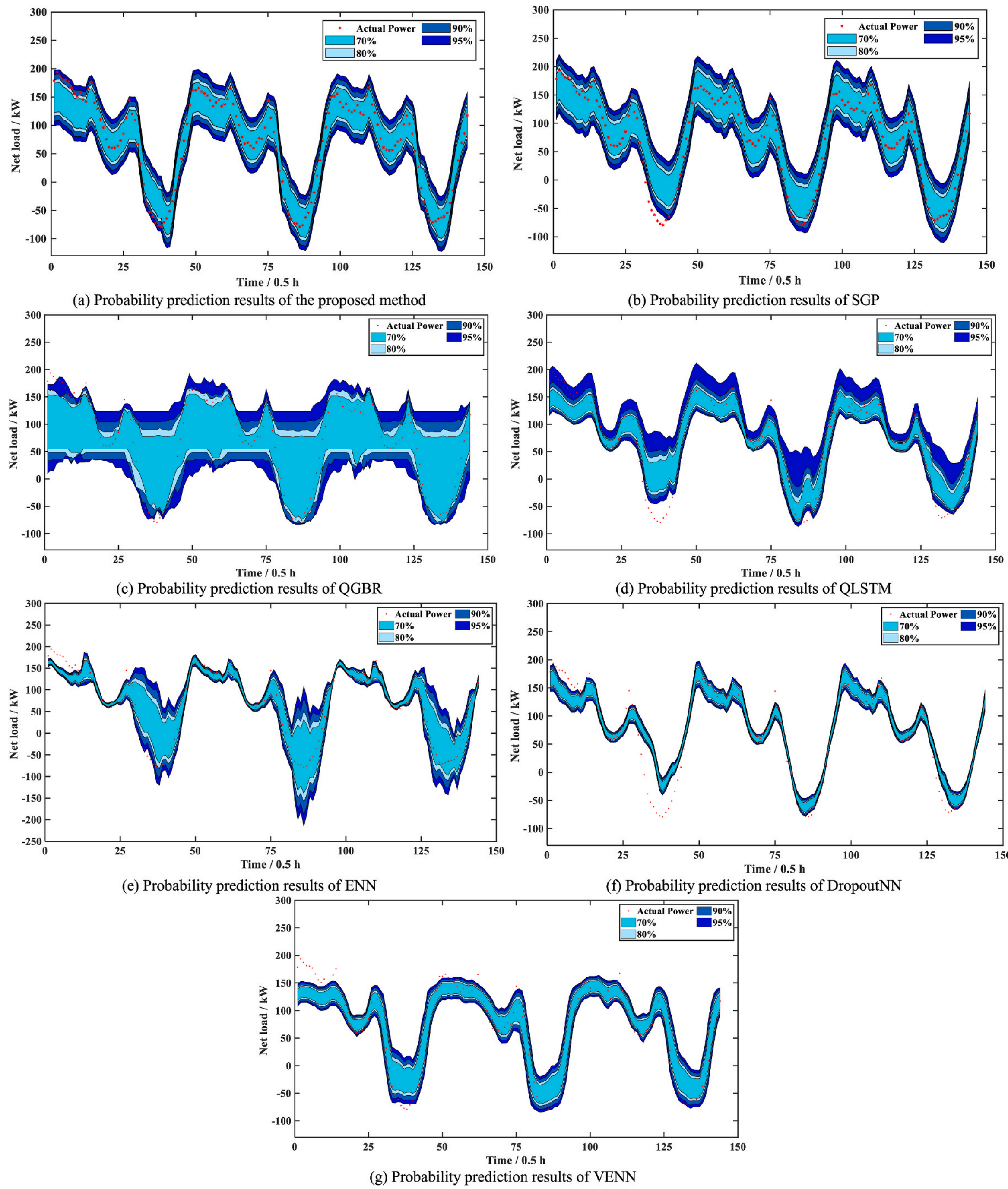
The forecasting intervals of different methods of the 3 days in the test dataset of Case 5 at 12 h task are plotted in Fig. 4 when the prediction intervals are set to 70%, 80%, 90%, 95%, respectively. The outcomes for CNN, LSTM, SVR, and PM are not shown as they are only point forecasting methods. It can be observed that the actual net load of some valley values cannot be covered by the 95% prediction interval of the SGP and QGBR methods, seeing t = 25–40 in Fig. 4(b) and t = 1–20 in Fig. 4(c) for example, which illustrate their insufficient reliability for net load forecasting. When DropoutNN is employed for probabilistic forecasting, its prediction intervals are very narrow so that they cannot cover the actual values, seeing t = 25–40 in Fig. 4(f) for example, which is consistent with the results in Tables 8–10. When the ENN method is constructed for net load probabilistic prediction, its prediction intervals are very wide at the valley of net load. It may be due to that the ENN method can hardly tackle the relatively larger uncertainty introduced by PV generation compared with traditional load. The negative impact of PV generation to the forecasting models can be observed here. By contrast, the proposed method constructs a Transformer model to extract key features and then obtain the probability outcomes through residual learning. This allows it to produce narrower prediction intervals that cover most of the actual values at the valley of the curve, which indicates that the sharpness and resolution of the proposed method outperform other probabilistic methods. It may be owing to that the Transformer forecasting model has better performance at the valley of the net load. In this case, the proposed method can incorporate the performance advantages of the Transformer and GP model, contributing to an effective uncertainty quantification for the net load, which is easy to implement for operators.

The point forecasting value of various methods and the probability density curve while the net load power is at curve valley and peak in Case 5 at 12 h task are shown in Fig. 5 to further illustrate the uncertainty capture ability of the proposed method. The outputs of QGBR and QLSTM are converted into Gaussian distribution, which is easy to compare with other methods. The prediction results at the load peak (t = 7:30 on 5th November) and at the load valley (t = 14:00 on 4th November) are shown in Fig. 5(a) and (b). It can be seen that the CNN, LSTM, and SVR methods can just provide exact point forecasting outcomes. By contrast, the QGBR, QLSTM, ENN, DropoutNN, and VENN can provide more useful uncertainty information. Among them, the prediction intervals of the QGBR method are very wide, resulting in less application value. Although the QLSTM method can achieve a relatively narrower prediction interval at the load peak, its distribution function is relatively far from the actual net load value compared with the proposed method. When the SGP method is employed to estimate the uncertainty of net load, its probability density curve is less steep than the proposed method at both the load peak and load valley. It may be owing to that the limited feature extraction capability of SGP cannot support it to get more confident results due to the stronger uncertainties at the valley and peak values. In this case, the SGP model tends to increase the prediction variance at these time points. By contrast, the proposed RML method quantifies the uncertainties of the net load based on the Transformer network-based forecasting model, which enables the proposed model to obtain more confident results and more reliable prediction intervals. The results demonstrate the superiority of the proposed residual modeling-based probabilistic forecasting method.

4.6. Analysis of the impact for dataset volume

To exclude the accidental factors of the dataset, the training datasets with different data volumes are used for the forecasting task. The models' structure and training process are the same for different data volumes. The task is a one-step forecasting task to predict the net load values in 12 h. Five dataset volume cases are considered to simulate various seasons, including 1) the training dataset at 200th-300th days and the testing dataset at 300th-310th 2) the training dataset at 210th-300th days and the testing dataset at 300th-310th 3) the training dataset at 220th-300th days and the testing dataset at 300th-310th 4) the training dataset at 230th-300th days and the testing dataset at 300th-310th 5) the training dataset at 240th-300th days and the testing dataset at 300th-310th. The comparative point forecasting methods include: 1) CNN; 2) LSTM; 3) Transformer network; 4) GBR; 4) SVR; 5) PM, where the actual values at the same time in the past day are directly considered as the predicted values. For the comparative probabilistic forecast methods, six cases are investigated, including 1) QGBR; 2) SGP; 3) QLSTM; 4) ENN; 5) DropoutNN; 6) VENN methods.

The MAE and RMSE achieved by various forecasting methods under different data volumes at 12 h forecasting task are listed in Table 11,

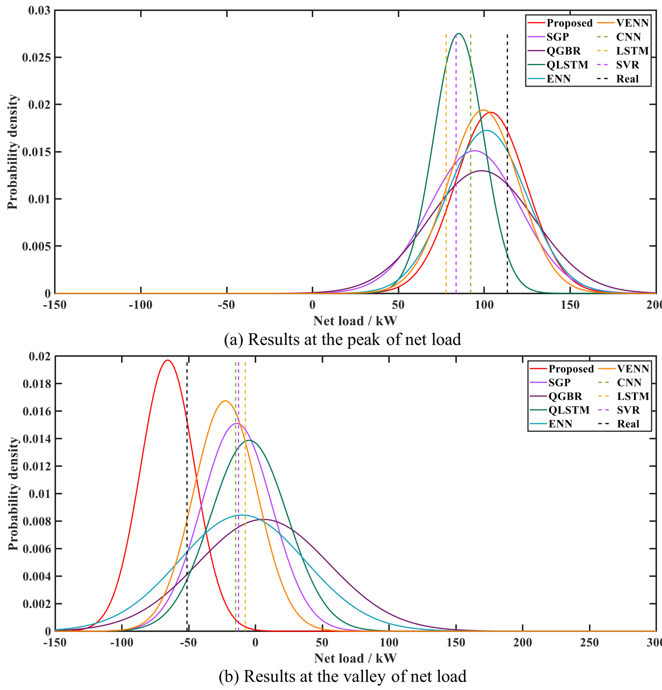


(a) Probability prediction results of the proposed method
(b) Probability prediction results of SGP
(c) Probability prediction results of QGBR
(d) Probability prediction results of QLSTM
(e) Probability prediction results of ENN
(f) Probability prediction results of DropoutNN
(g) Probability prediction results of VENN

Fig. 4. Probability prediction results of different methods.

where the best results are bold. It can be seen from the table that the prediction performance of the LSTM and GBR methods fluctuates since the process of model construction and training is inevitably affected by epistemic uncertainty. Therefore, it is difficult for such forecasting methods to obtain the ideal optimal situation. By contrast, the proposed method is insensitive to different dataset sizes, seeing its MAE values under 60–100 days data volume. It is achieved by building a GP to further learn the forecasting tasks based on the existing Transformer model without changing the existing construction and training process of the original model, which also makes the proposed RML method applicable and easy to implement.

Comparative tests are carried out among different probabilistic methods to verify the uncertainty quantification capability of the proposed RML model. The CRPS, Pinball loss, and Winkler score ($\alpha = 0.1$) achieved by various forecasting methods under different data volumes at 12 h forecasting task are listed in Table 12, where the best results are bold. It can be observed from the table that the QGBR suffers larger prediction errors due to insufficient ability to learn correlation from time series. Although the neural network-based methods, such as QLSTM, ENN, and DropoutNN, can achieve better probabilistic outcomes due to the stronger feature extraction capability, their prediction performance fluctuates at various training dataset conditions. By contrast, the prediction performance of the SGP method can achieve more stable probabilistic outcomes facing various training datasets due to its small sample learning characteristics. In this case, the proposed method incorporates the advantages of neural network-based and GP



(a) Results at the peak of net load

(b) Results at the valley of net load

Fig. 5. Probabilistic density results and point forecasting results of various methods.

Table

11MAE and RMSE achieved by various forecasting methods under different data volumes at 12 h forecasting task.

Metrics	Models	Training dataset size (days)				
		100	90	80	70	60
MAE (kW)	Proposed	**15.4**	**15.2**	**15.3**	**15.1**	**15.2**
	CNN	16.9	16.4	16.1	16.4	16.4
	LSTM	16.5	17.7	17.8	17.3	17.1
	Transformer	16.0	15.8	16.5	16.1	15.7
	GBR	17.1	16.5	16.5	16.8	15.4
	SVR	18.6	19.0	19.1	19.0	19.8
	PM	21.6	21.6	21.6	21.6	21.6
RMSE (kW)	Proposed	**21.4**	**21.4**	**21.1**	**21.4**	**21.2**
	CNN	25.2	23.7	23.2	24.4	23.7
	LSTM	22.9	24.2	25.3	25.6	23.9
	Transformer	22.7	21.9	23.2	23.8	21.9
	GBR	24.2	24.0	24.4	24.6	22.1
	SVR	24.5	25.0	25.1	25.0	25.7
	PM	29.3	29.3	29.3	29.3	29.3

methods. A transformer network is first employed to learn the pattern of net load, followed by a composite GP kernel to capture the uncertainty of the net load and generate the probabilistic distribution function. This allows the proposed method to stably outperform other methods at different dataset volumes. The results further demonstrate the superiority of the proposed residual modeling-based probabilistic forecasting method.

4.7. Ablation experiments for the proposed RML framework

More tests are carried out to verify the performance of the proposed RML method. The task is a one-step forecasting task to predict the net load values in 30 min. Five dataset cases are considered to simulate various seasons, including Case 1) the training dataset at 0th-100th days and the testing dataset at 100th-110th; Case 2) the training dataset at 50th-150th days and the testing dataset at 150th-160th; Case 3) the training dataset at 100th-200th days and the testing dataset at 200th-210th; Case 4) the training dataset at 150th-250th days and the testing dataset at 250th-260th; Case 5) the training dataset at 200th-300th days and the testing dataset at 300th-310th. The comparative point forecasting methods include: 1) LSTM; 2) LSTM-RML, where the proposed RML method is used to revise the original LSTM model; 3) Transformer network; 4) PM, where the latest actual values are directly considered as the predicted values. For the comparative probabilistic forecast methods, three cases are investigated, including 1) SGP; 2) QLSTM; 3) LSTM-RML, where the proposed RML method is used to quantify the uncertainty and convert the point forecasting results to probabilistic information.

Table

12CRPS, Pinball loss, and Winkler score ($\alpha = 0.1$) achieved by various forecasting methods under different data volumes at 12 h forecasting task.

Metrics	Models	Training dataset size (days)				
		100	90	80	70	60
CRPS (kW)	Proposed	**12.12**	**12.29**	**11.89**	**12.16**	**12.32**
	QGBR	16.40	16.60	16.56	16.65	16.56
	SGP	13.09	13.01	13.19	13.19	13.59
	QLSTM	13.84	14.25	14.24	14.50	15.40
	ENN	12.80	13.03	12.83	14.08	14.97
	DropoutNN	14.43	13.82	14.71	15.68	15.19
	VENN	14.36	14.65	14.06	14.92	14.97
Pinball (kW)	Proposed	**6.106**	**6.186**	**5.992**	**6.123**	**6.203**
	QGBR	8.253	8.350	8.331	8.376	8.327
	SGP	6.593	6.554	6.646	6.643	6.850
	QLSTM	7.007	7.207	7.182	7.336	7.768
	ENN	6.461	6.556	6.459	7.072	7.506
	DropoutNN	7.226	6.968	7.410	7.904	7.652
	VENN	7.257	7.407	7.116	7.555	7.571
Winkler (kW) (α = 0.1)	Proposed	**103.4**	**107.2**	**99.77**	**104.3**	**106.7**
	QGBR	143.3	145.3	141.1	148.5	152.6
	SGP	118.5	118.1	117.8	118.5	120.2
	QLSTM	120.0	114.8	123.7	142.5	138.9
	ENN	125.5	124.9	122.6	145.9	159.6
	DropoutNN	196.3	200.0	220.7	228.1	229.1
	VENN	138.8	144.1	124.0	113.8	149.0

Table

13MAE and RMSE achieved by various forecasting methods in ablation experiments at 30 min forecasting task.

Metrics	Models	Training dataset positions				
		Case 1	Case 2	Case 3	Case 4	Case 5
MAE (kW)	Proposed	**7.34**	**7.22**	**8.17**	**6.37**	**7.64**
	LSTM	11.3	8.49	11.3	8.20	10.9
	LSTM-RML	10.1	8.33	10.9	8.00	9.13
	Transformer	9.66	7.80	8.59	6.49	8.64
	PM	13.1	10.9	9.84	11.7	13.6
RMSE (kW)	Proposed	**9.92**	**9.56**	**11.5**	**8.58**	**10.3**
	LSTM	15.4	11.3	17.1	12.8	16.3
	LSTM-RML	13.4	11.0	16.3	11.4	12.0
	Transformer	12.8	10.4	12.1	8.73	11.5
	PM	17.0	14.2	12.9	15.2	16.9

The MAE and RMSE achieved by various forecasting methods at 30 min forecasting task are listed in Table 13, where the best results are bold. When the proposed RML method is implemented behind the original LSTM model, an obvious performance improvement can be observed in the LSTM-RML method under different dataset conditions. For example, the MAE performance of LSTM-RML increases by 10.6% under Case 1 and 16.3% under Case 5 compared with the original LSTM model. The results demonstrate that the RML method can be implemented for the existing point forecasting models and further improve their forecasting performance, which indicates its significance in real applications. Besides this, although the LSTM-RML method can achieve several improvements compared with the original LSTM method, its forecasting results are still worse than the proposed method. It may be due to that the designed Transformer network can yield more precise forecasting outcomes, which can guide the RML to learn more clear and

Table 14
CRPS, Pinball loss, and Winkler score (α = 0.1) achieved by various forecasting methods in ablation experiments at 30 min forecasting task.

Metrics	Models	Training dataset positions				
		Case 1	Case 2	Case 3	Case 4	Case 5
CRPS (kW)	Proposed	**5.517**	**5.350**	**6.164**	**4.727**	**5.690**
	SGP	6.466	6.549	6.201	6.247	6.836
	QLSTM	11.97	9.044	8.715	7.092	7.934
	LSTM-RML	7.491	6.134	8.505	6.074	6.696
Pinball (kW)	Proposed	**2.783**	**2.699**	**3.110**	**2.387**	**2.874**
	SGP	3.259	3.304	3.132	3.152	3.452
	QLSTM	6.050	4.573	4.407	3.591	4.018
	LSTM-RML	3.786	3.090	4.296	3.063	3.382
Winkler (kW)	Proposed	**45.24**	**41.99**	**49.71**	**39.36**	**47.90**
(α = 0.1)	SGP	51.32	49.57	57.40	52.24	52.13
	QLSTM	137.5	95.37	99.62	68.22	77.98
	LSTM-RML	61.67	49.88	88.34	53.25	53.83

regular residual signals. The experiment results also indicate the superiority of the proposed Transformer structure-based forecasting model.

Comparative tests are carried out among different probabilistic methods to verify the uncertainty quantification capability of the proposed RML model. The CRPS, Pinball loss, and Winkler score (α = 0.1) obtained by various methods at 30 min forecasting task are listed in Table 14, where the best results are bold. It can be observed from the table that, without constructing numerous quantiles or training numerous models such as ensemble-based methods, the RML method can directly update the original point forecasting model and further provide it with the probabilistic prediction capability, seeing the LSTM-RML method for example. Its probabilistic forecasting performance is even better than the traditional QLSTM method with an obvious margin under most dataset conditions without changing the original model construction and introducing more training processes. The experiments further demonstrate that the proposed RML method has effective extendibility and is easy to implement for existing forecasting models in real applications, such as upgrading existing point forecasting models by converting them into probabilistic predictions or adapting them for new usage scenarios. Besides this, the forecasting performance of LSTM-RML outperforms the traditional SGP method owing to the stronger feature extraction from the LSTM model, seeing their Pinball loss under Cases 2, 4, and 5. The results indicate the effectiveness of the proposed method in incorporating the advantages of neural networks and GP methods.

5. Conclusion

This paper designs a probabilistic net load forecasting framework based on the Transformer network and RML process. Comparative tests with various point forecasting and probabilistic forecasting methods utilizing real-world historical data illustrate that: 1) the proposed Transformer structured-based model can achieve better point forecasting results than benchmarking methods. The performance achieved by the proposed method outperforms other methods, especially at the valley of the net load; 2) the proposed approach can more effectively capture the uncertainty of net load than the benchmarking methods. It has higher reliability and more effective prediction intervals; 3) the proposed RML method can improve the accuracy of the original forecasting model without changing the model structure and training process, which indicates that the proposed RML method can solve the epistemic uncertainty to some extent and optimize the original forecasting models.

CRediT authorship contribution statement

Jiaxiang Hu: Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Weihao Hu:** Funding acquisition, Investigation, Supervision. **Di Cao:** Supervision, Writing – review & editing. **Xinwu Sun:** Investigation. **Jianjun Chen:** Investigation, Visualization. **Yuehui Huang:** Investigation. **Zhe Chen:** Investigation, Supervision. **Frede Blaabjerg:** Investigation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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